

Lab: 10

perform compression on MNIST Dataset using Autoencoders

Aim:

To implement an autoencoder for compressing and reconstructing images from the MNIST dataset.

Objectives:

1. To understand the concept and working of autoencoder.
2. To perform image compression using the encoder-decoder arch.
3. To visualize and analyze reconstructed image and their quality.
4. To explore dimensionality reduction through neural network.

Pseudocode:

Start

1. Import necessary libraries (tensorflow/tensor, numpy)
2. Load MNIST dataset.

3. Define Autoencoder arch

encoder:

-> input layer

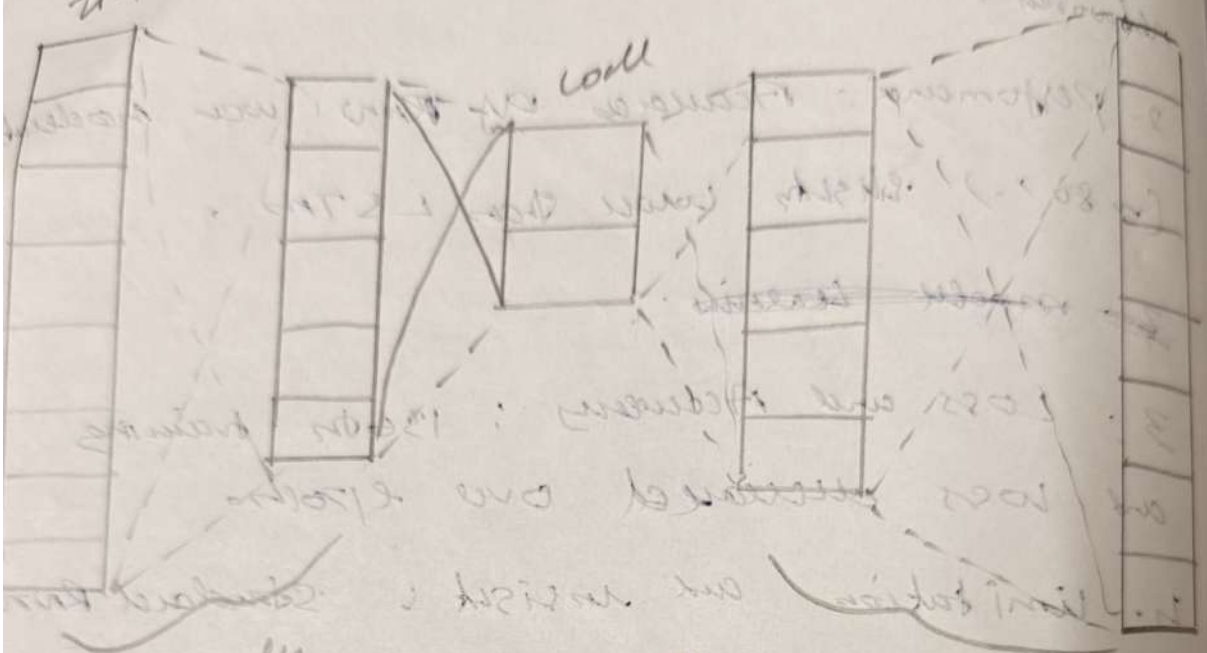
-> Dense (128, activation = 'relu')

decoder:

-> Dense (64, activation = 'relu')

4. Compile the model:

-> loss: mean squared error (mse)



When checking with long features.
 middle
 various percent
 errors

① Encodings implemented as recurrent neural network. to capture sequential data.

5. Train model using MNIST training data
6. Evaluate model performance on test data.
7. Visualize:
 - original vs Reconstructed image
 - plot

End

Observation:

1. Training Behaviour:

- + The autoencoder successfully learned to reconstruct digit 1max.

2. Compression effect.

- + The encoder reduced 784-dimensional input into a 32-dimensional latent space.

3. Visualization:

- + plots showed strong similarity between original and reconstructed digits.

4. Interpretation:

- + The model efficiently performed non-linear dimensionality reduction, PCA which is linear.

Result:

Successfully implemented an Autoencoder for compression of MNIST image.

